Unless explicitly requested by a problem, do not include sketches as part of your proof. You are free to use the result from any problem on this (or previous) assignment as a part of your solution to a different problem even if you have not solved the former problem.

Problem 1 (2pts)**.** Prove that $R(3,n) \leqslant n^2$ for every positive integer n.

Solution. Since $n+1 \leqslant 2n$, we have

$$R(3,n) \leqslant \binom{n+3-2}{3-1} = \binom{n+1}{2} = \frac{n(n+1)}{2} \leqslant n^2.$$

□

Problem 2 (2pts)**.** The 3-color Ramsey number $R(m,n,p)$ is the smallest integer N such that every 3-coloring of $E(K_N)$ (say with colors red, blue and green) contains either a red copy of K_m, a blue copy of K_n or a green copy of K_p.

Prove that $R(3,3,3) \leqslant 17$.

Solution. Fix any red,blue,green-coloring of $E(K_{17})$ and fix any vertex $v \in V(K_{17})$. Let R, B, G denote the red, blue and green neighborhoods of v, respectively, so $N(v) = R \sqcup B \sqcup G$. In particular, $16 = |R|+|B|+|G|$. If all three of these sets had size at most 5, then we would have $absR+|B|+|G| \leqslant 15$, a contradiction. Thus, one of these neighborhoods must have size at least 6. Since we are looking for the same structure in each color (namely, a triangle), we may suppose that $|R| \geqslant 6$ without loss of generality.

Let $G \subseteq K_{17}$ denote the subgraph induced by R, and consider restricting the coloring to G. If there is any edge $xy \in E(G)$ which is colored red, then $\{v,x,y\}$ would form a red triangle and we would be done. Otherwise, every edge in G is either blue or green. Then, since $|R| \geqslant 6 = R(3,3)$, we know that this restricted coloring contains either a blue triangle or a green triangle, as needed. □

Problem 3 (2pts)**.** Suppose that G is a graph that has no induced $K_{1,4}$ and such that $\omega(G) = 4$. Show that $\Delta(G) \leqslant 17$.

Hint: $17+1 = R(4,4)$.

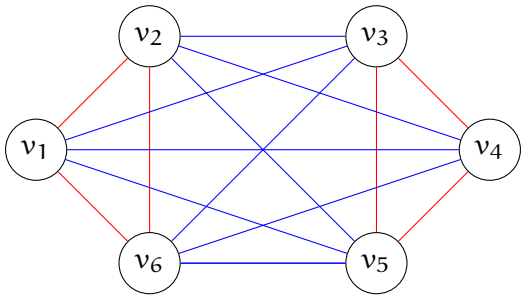
Solution. Suppose for the sake of contradiction that $\Delta(G) \geqslant 18$. Then some $v \in V(G)$ has $\deg v \geqslant 18$. Let $H \subseteq G$ denote the subgraph induced by $N(v)$.

Since H is an 18-vertex graph and $18 = R(4,4)$, we know from class that either $\alpha(H) \geqslant 4$ or $\omega(H) \geqslant 4$ (if you don't like this interpretation of $R(4,4)$, consider a red,blue-coloring of K_{18} where the edges in H are red and the edges in $\overline{H}$ are blue). Thus either H contains a size 4 independent set or H contains a K_4. In the former, the 4 independent vertices of H together with v form an induced $K_{1,4}$ in G, a contradiction. In the latter case, the 4 vertices in the $K_4 \subseteq H$ together with v form a K_5 in G, which contradicts $\omega(G) = 4$. □

Problem 4 (2pts)**.** Show that $N = 7$ is the smallest integer N such that every red,blue-coloring of $E(K_N)$ contains either a red $K_{1,3}$ or a blue K_3.

Solution. Let N denote the smallest integer in question.

First, we show that $N \geqslant 7$. Indeed, for $E(K_6)$ we have the following red,blue-coloring:



Since the red graph is two disjoint copies of K_3, there is no red $K_{1,3}$, which is connected on 4 vertices. Since the blue graph is $K_{3,3}$, which has chromatic number 2, there is no blue K_3.

Second, we show that $N \leqslant 7$. Assume for a contradiction that there is a red,blue- coloring of $E(K_7)$ that produces neither a red $K_{1,3}$ nor a blue K_3. Fix any vertex v in K_7 and consider the partition $N(v) = R \sqcup B$ into the red and blue neighborhoods of v. We have $6 = |R| + |B|$. Since the coloring contains no red $K_{1,3}$, we have $|R| \leqslant 2$, thus $|B| \geqslant 4$. Now we consider two cases.

Case 1: there are x, y in B such that the edge xy is blue. In this case, $\{v, x, y\}$ form a blue K_3, contradiction.

Case 2: for all x, y in B, the edge xy is red. In this case, since $|B| \geqslant 4$, within B there is a red K_4, so in particular a red $K_{1,3}$, contradiction.

In any case we get a contradiction, thus any red,blue- coloring of $E(K_7)$ produces either a red $K_{1,3}$ or a blue K_3. □

Problem 5 (2 pts)**.** Let n be a positive integer and set $N = 3n - 1$. Construct a red,blue-coloring of $E(K_{N-1})$ which *does not* contain a monochromatic matching with n edges.

Hint: Break $V(K_{N-1})$ into two pieces, one of size $2n - 1$ and the other of size $n - 1$, and color the edges based on how they intersect these two pieces.

Solution. Since $N = 3n - 1$, we can partition $V(K_{N-1}) = U \sqcup W$ where $|U| = 2n - 1$ and $|W| = n - 1$. Color every edge with both end-points in A red, and color every other edge blue. We claim that this coloring has no monochromatic matching with n edges.

Indeed, suppose for the sake of contradiction that M were a monochromatic matching with n edges.

If M is red, then every end-point of an edge in M must reside in A. Since M is a matching, all of these end-points are distinct and so we must have $|A| \geqslant 2n$, which is not the case.

If M is blue, then every edge in M must have at least one end-point in B (since otherwise that edge would be red). But M has n many edges and these edges are vertex-disjoint, so this would only be possible if $|B| \geqslant n$, which is not the case. □